SC306 Homework 5

Othar Zaldastani II

March 17, 2026

Part 1: AR and PAR

A)

$$AR\ \% = \frac{R_e - R_u}{R_e} \times 100$$

```
## [1] 77.1
```

The attributable risk percent of low birth weight from Cocaine use is 77.1%.

B) This means approximately 77% of the low birth weight risk for infants born to mothers who used Cocaine while pregnant is attributable to their use of Cocaine.

C)

$$PAR\ \% = \frac{P_e(R_e - R_u)}{P_e R_e + (1 - P_e)R_u} \times 100$$

```
## [1] 6.3
```

The population attributable risk percent of low birth weight from Cocaine use is 6.3%.

D) Approximately 6.3% of the low birth weight risk for all infants born in California is attributable to the use of Cocaine while pregnant.

Part 2: Study Designs

Abstract 1

The researchers used a randomized control trial (RCT) with a six-month follow up.

Effects of reduction in screen use/exposure via the use of a modified classroom curriculum.

Childhood obesity related measurements such as BMI and bodily dimensions/ratios.

The primary measures used to quantify disease were obesity-related measures such as BMI and body measurements. Since a condition like obesity follows more of a sliding scale of severity, changes in bodily characteristics were used as opposed to number of participants over/under a threshold.

The measures used to quantify exposure vs. outcome relationship was difference in means (difference in differences) with 95% confidence intervals.

Abstract 2

The researchers used a cohort study with continuous, indefinite follow ups to conduct this experiment. The cohort was followed from the birth period of 1915-1929 to 1995 and mortality causes were assessed.

The exposure investigated was fetal growth rate (controlling for size at birth).

The disease investigated was mortality from Ischaemic Heart Disease (and other causes) during the lives of the participants.

The measures used to quantify disease were mortality rates from Ischaemic Heart Disease and other Cardiovascular diseases.

The measure used to quantify the relationship between exposure and outcome was incidence rate ratios relative to birth weight with 95% confidence intervals. An additional set of results was published controlling for socioeconomic factors at birth and adulthood.

Abstract 3

The researchers used a case-control study. Women with premenopausal breast cancer were compared to demographically-matched controls. Prior history of electric blanket use was assessed retrospectively.

The exposure investigated was electric blanket use (as a proxy for increased background exposure to 60 hz electromagnetic fields). Data on blanket use included time, duration, and frequency of use over the previous decade.

The disease investigated was premenopausal breast cancer.

Since this is a case-control study, disease frequency is not a meaningful statistic. Numbers of cases and controls were chosen at the start of the study.

The measure used to quantify relationship between electric blanket use and premenopausal breast cancer are odds ratios with 95% confidence intervals. The odds ratios were adjusted for age, education, and other possible risk factors/confounding variables.

Part 3: A Case-Control Study

A) Since cervical cancer is relatively uncommon, it makes sense to do a case-control study compared to other options such as an RCT or cohort study. An RCT would be challenging and extremely unethical, while a cohort study would require an immense number of participants and time in order to have a sufficient number of female participants develop cervical cancer. A case-control study is best because the disease takes a long time to incubate, and the retrospective information needed to categorize participants is not very granular (number of sexual partners, sex worker visitation, etc.) so there is a lower likelihood of receiving false information.

B) Some potential problems include recall bias, reporting bias, selection bias, and confounding issues. Cases and their husbands may put more effort into recalling past information compared to controls. Contact with sex workers is stigmatized and some husbands may lie about their history. The selection of individuals from hospitals also poses a potential problem because hospitalized individuals are not a representative segment of the population. Also, some potential confounding variables include exposure to carcinogens such as cigarette use, socioeconomic status, occupation or occupational hazard and exposure, and the wives' number of sexual partners prior to (or during) marriage.

C) 2x2 table of cervical cancer data

	cancer	No cancer
CSW	179	518
No CSW	42	267

D) Women with husbands who had previously visited sex workers had higher odds of cervical cancer compared to women with husbands who had reported never having visited a sex worker.

E) The prevalence of cervical cancer among study participants is 22%.

$$\frac{\text{\# of Cases in Study}}{\text{\# of Participants in Study}} = \frac{179 + 42}{179 + 518 + 42 + 267} = 0.220$$

```
## [1] 0.22
```

F) No, because in a case-cohort study, participants are not randomly selected but, so the study population has a significantly higher rate of the condition (here, cervical cancer) than the general population. Investigators intentionally set the number of cases and controls in the study.

G) The 95% confidence interval for the odds ratio is [1.53 : 3.2] centered at 2.19.

```
## estimate    lower    upper
##     2.19     1.53     3.20
```

H) The odds ratio of 2.19 with a 95% confidence interval of [1.53 : 3.2] means that the odds of developing invasive cervical cancer among women with husbands who had previously visited sex workers is estimated to be between 1.5 and 3.2 times the odds of the general population with husbands who did not visit sex workers.

Code Appendix

```r
knitr::opts_chunk$set(          # setup chunk
  echo = FALSE,
  warning = FALSE,
  message = FALSE,
  cache = FALSE,
  fig.align = "center",
  attr.source = '.numberLines',
  cache = T)

library(tidyverse)              # loading packages
library(dplyr)
library(knitr)
library(epitools)

set.seed(24)                    # seed for reproducibility

kable2 <- function(x, ...) {    # for debugging table creation
  kableExtra::kable_styling(
    knitr::kable(x, ...))}
# Part 1
# Question A
Re = (110)/(110+289)            # Risk among exposed
Ru = (1073)/(1073+15896)        # Risk among unexposed
AR = (Re - Ru)/(Re) * 100       # Attributable risk
round(AR, 1)

# Question C
Pe = 0.02                       # Exposure prevalence
PAR = (Pe * (Re - Ru)) / (Pe * Re + (1-Pe) * Ru) * 100
round(PAR, 2)

# Part 3
# Question C
table.cancer =                  # Creation of 2x2 table
  matrix(c(179, 518, 42, 267),nrow=2, byrow = TRUE)
colnames(table.cancer) = c('cancer', 'No cancer')
rownames(table.cancer) = c('CSW', 'No CSW')
kable(table.cancer)             # Printing 2x2 table

# Question E
round(((179+42)/(179 + 518 + 42 + 267)), 3)
# Question G
or = oddsratio(table.cancer, rev = "b")
round(or$measure["CSW", ], 2)
```

Attributions: I have used some template/styling code from GitHub user alexpghayes.